

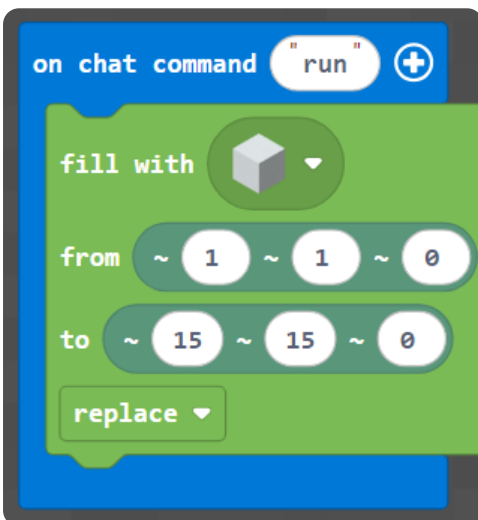
3D Coordinates · Pixel Art on the Wall (x, y) — English Worksheet (Beginner)

Topic: Pixel Art on the Wall (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This lesson you copy a picture made of blocks onto your wall. Each block sits at **(x, y)** — x is **across**, y is **up**.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your home spot:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too.

1 · Predict 🤖

Read the steps. Circle your answer.

```
place red block at (6, 1)
place red block at (7, 1)
place red block at (8, 1)
```

Across or up? Circle one: across · up

Which number changes? Circle one: x · y

```
place green block at (8, 13)
place green block at (8, 14)
```

Across or up? Circle one: across · up

Which number changes? Circle one: $x \cdot y$

2 · Spot the Bug 🐛

This should make a row going **across**: (6, 1), (7, 1), (8, 1). But all three blocks land in the **same spot**.

```
place yellow block at (6, 1)
place yellow block at (6, 1)
place yellow block at (6, 1)
```

Hint: to go across, the **x** must grow. Fill in the blanks.

```
place yellow block at (6, 1)
place yellow block at (__ , 1)
place yellow block at (__ , 1)
```

Which two numbers are missing? Circle one: 7 and 8 · 6 and 6

Why was it broken? Circle one: all three had the same x · the color was wrong

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block).

Warm up first. Copy this small green cross. Read each green square's (x, y) and place a green block there.

■ Green cross

	1	2	3	4	5
5			■		
4			■		
3	■	■	■	■	■
2			■		
1			■		

Try a big one. Pick the 🍍 pineapple or the 🍏 apple from the main worksheet grid and copy it the same way. Read each colored square and place that color at its (x, y).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.